

Introduction to Analytics

Turning Data into Insight. Insight into Action. Action into Value.

Analytics is the process of collecting, transforming, and organizing data to identify patterns, draw conclusions, and make informed decisions. This lesson explores what analytics is, the types of analytics, key patterns, tools, and how it relates to data analysis and data science.

“Data is the new oil, but analytics is the engine that turns it into power.”
— Unknown

THE ANALYTICS LANDSCAPE

Analytics is not a single tool or activity—it’s a journey from raw data to better decisions.



COMMON PATTERNS IDENTIFIED IN ANALYTICS

Data can reveal many types of patterns. Recognizing them helps us understand the past and shape the future.

1. Trend Analysis

- Upward trend ↗
- Downward trend ↘
- Horizontal trend →

Example: Sales increasing quarter over quarter.

2. Seasonal & Cyclical

- Seasonality:** repeats over fixed intervals (day, week, month, year).
- Cyclical:** longer, less predictable cycles.

Example: Retail sales spike in holiday season (seasonality).

3. Correlation & Relationships

- Positive:** variables move in same direction.
- Negative:** variables move in opposite direction.
- No correlation:** no pattern.

Example: Temp ↑ → Ice cream sales ↑

4. Clustering & Segmentation

- Group similar data points together.
- Enables market segmentation and behavioral grouping.

Example: Group customers by purchase behavior.

5. Anomalies & Outliers

- Data points that deviate significantly.
- Useful for fraud detection and system monitoring.

Example: Unusual credit card transaction amount.

6. Association & Frequent Itemsets

- Finds items that frequently occur together.
- Used in market basket analysis and web usage mining.

Example: Customers who buy milk also buy bread.

HOW ANALYTICS IDENTIFIES PATTERNS

- Data Visualization**
Charts, graphs, and plots reveal trends, correlations, and anomalies visually.
- Statistical Analysis**
Summarizes data and finds relationships (averages, regression, etc.).
- Machine Learning**
Algorithms learn from data to classify, predict, and cluster automatically.
- Data Mining**
Discovers hidden patterns and relationships in large datasets.

THE 4 TYPES OF ANALYTICS

Descriptive
What happened?
Summarizes past data to understand events.

Diagnostic
Why did it happen?
Investigates root causes behind past events.

Predictive
What will happen?
Uses historical data to forecast future outcomes.

Prescriptive
What should we do?
Recommends actions to achieve desired outcomes.

TOOLS I KNOW

- Microsoft Excel & Access
- Python & R
- Power BI & Tableau
- Minitab (Statistical Analysis)

DATA ORGANIZATION AND TRANSFORMATION

Structured Data
(SQL Database, Wonderware, PostgreSQL)

Organized in rows and columns.
Easy to store, query, and analyze.

Unstructured Data
(MongoDB)

No predefined format.
Requires special tools to process.

Big Data
(APACHE SPARK, APACHE HADOOP, etc.)

Large volume, velocity, variety.
Requires distributed processing.

HOW IS ANALYSIS DIFFERENT FROM ANALYTICS?

- Analysis is a subset of analytics.**
With analysis, you have the data in front of you—you don’t really care how it got there or where it came from. Analysis is about using the tools and techniques at your disposal to get meaning and actionable insight from the dataset.
- Data science is about using data to tell stories.**
But before they can narrate the stories, data scientists have to cleanse the data, explore it, and build models to discover and communicate those stories.



AT A GLANCE

- Analytics** answers questions and drives decisions.
- Analysis** is the act of examining data.
- Data Science** builds the models and stories that explain and predict.